MSG No. 135.484 $P4_2/mbc1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m..1'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,6,9,14,17,22,25,30]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,3,10,11,18,19,26,27]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,5,12,13,20,21,28,29]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,15,16,23,24,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,6,10,11,17,22,26,27]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[2,3,9,14,18,19,25,30]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,5,15,16,20,21,31,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[7,8,12,13,23,24,28,29]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,6,9,14,17,22,25,30]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,3,10,11,18,19,26,27]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[4,5,12,13,20,21,28,29]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,15,16,23,24,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.221'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6,7,8,17,22,23,24]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[2,3,4,5,18,19,20,21]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,14,15,16,25,30,31,32]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[10,11,12,13,26,27,28,29]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..1'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,6,17,22]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,3,18,19]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5,20,21]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,8,23,24]`
5	$[0,\ 0,\ -z]$	`[9,14,25,30]`
6	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[10,11,26,27]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[12,13,28,29]`

8	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,16,31,32]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,17,22]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3,18,19]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[4,5,20,21]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,8,23,24]`
5	$\left[0,\ \frac{1}{2},\ -z\right]$	`[9,14,25,30]`
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[10,11,26,27]`
7	$\left[\frac{1}{2},\ 0,\ z\right]$	`[12,13,28,29]`
8	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,16,31,32]`

Table 7: Wyckoff site: `8g`, site symmetry: `..21'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[1,7,17,23]`
2	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	`[2,5,18,21]`
3	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	`[3,4,19,20]`
4	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[6,8,22,24]`
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[9,15,25,31]`
6	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	`[10,13,26,29]`
7	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	`[11,12,27,28]`
8	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[14,16,30,32]`

Table 8: Wyckoff site: `8h`, site symmetry: `m..1'`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,14,17,30]`
2	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[2,11,18,27]`
3	$\left[y,\ -x,\ \frac{1}{2}\right]$	`[3,10,19,26]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[4,13,20,29]`
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	`[5,12,21,28]`
6	$[-x,\ -y,\ 0]$	`[6,9,22,25]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,16,23,32]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[8,15,24,31]`

Table 9: Wyckoff site: `16i`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,17]`
2	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	`[2,18]`
3	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	`[3,19]`

continued ...

Table 9

No.	position	mapping
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[4,20]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[5,21]
6	$[-x,\ -y,\ z]$	[6,22]
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[7,23]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[8,24]
9	$[-x,\ -y,\ -z]$	[9,25]
10	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	[10,26]
11	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	[11,27]
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[12,28]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[13,29]
14	$[x,\ y,\ -z]$	[14,30]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[15,31]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16,32]